

An Equal-Edge Counterexample to the Proposed Monotonicity of $\|T_G\|$

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Status. Candidate counterexample, likely valid. This packet answers negatively the edge-count monotonicity question raised in Section 6 of Santos–Tripathi–Youssef [1].

1 Source Question

For a graph G on L vertices, the source writes

$$T_G = \frac{1}{\sqrt{L}} \sum_{i=1}^L s_i,$$

where the s_i are G -independent semicircle variables. Section 6 asks whether the following refined monotonicity property holds: if G, G' are graphs on $[L]$ with the same clique number and $|E(G)| \leq |E(G')|$, must $\|T_G\| \leq \|T_{G'}\|$?

The relevant passage is on page 16 of [1].

satisfy $\text{hom}(F_\pi, G) \ll L^p$. Understanding how the structure of the partition influences $\text{hom}(F_\pi, G)$ is a promising direction for further study.

We also note that the bound $\|T_G\| \leq 2\sqrt{\chi(G)}$ follows from the monotonicity of the norm with respect to edge addition (see Remark 2.3). Since G is $\chi(G)$ -colorable, there exists a partition $I_1, \dots, I_{\chi(G)}$ of $[L]$ such that $E(G) \subseteq E(K_{I_1, \dots, I_{\chi(G)}})$, where $K_{I_1, \dots, I_d}$ denotes the complete d -partite graph over $I_1, \dots, I_d$. The corresponding norm satisfies

$$\left\| T_{K_{I_1, \dots, I_{\chi(G)}}} \right\| = 2 \sum_{k=1}^{\chi(G)} |I_k|^{1/2},$$

from which the bound follows via the Cauchy–Schwarz inequality.

As mentioned in the introduction, it would be desirable to develop a Turán-type approach to prove Corollary 1.2. Moreover, it is unclear whether a more refined monotonicity property holds. Specifically, given two graphs G, G' on $[L]$ with the same clique number and $|E(G)| \leq |E(G')|$, does it follow that $\|T_G\| \leq \|T_{G'}\|$? A positive answer would suggest a constructive route to identifying extremizers and potentially establish uniqueness. More broadly, a better understanding of which graph parameters control $\|T_G\|$, beyond the clique number, remains an open problem of independent interest.

Finally, we note that Haagerup-type inequalities [17, 22, 4, 23, 46], which generalize Khintchine-type bounds, offer a natural direction for extending our results. Adapting such inequalities to the graph convolution framework introduced here could lead to further structural insights.

REFERENCES

2 Idea of the Proof

The proposed monotonicity is already too strong at equal edge count. If two graphs have the same number of edges and the same clique number, the proposed statement applies in both directions and therefore predicts equality of $\|T_G\|$. Complete multipartite graphs give an explicit norm formula: the norm is proportional to the sum of square roots of the part sizes. Edge count fixes the sum of the squares of the part sizes, but it does not fix the sum of their square roots. Two complete 4-partite graphs on 10 vertices exploit exactly this gap.

3 Complete Multipartite Norm Formula

Lemma 1. *Let $K_{I_1, \dots, I_r}$ be the complete r -partite graph on $L = \sum_{k=1}^r |I_k|$ vertices, and let T_K be the normalized operator from the source paper. Then*

$$\|T_K\| = \frac{2}{\sqrt{L}} \sum_{k=1}^r \sqrt{|I_k|}.$$

Proof. In the complete multipartite graph, vertices in distinct parts are adjacent, so the corresponding algebras commute and are classically independent. Vertices in the same part are nonadjacent, so within each part the corresponding semicircle variables are free.

For each part set

$$X_k = \sum_{i \in I_k} s_i.$$

The free sum of $|I_k|$ standard semicircle variables is a semicircle variable of variance $|I_k|$, hence has spectrum $[-2\sqrt{|I_k|}, 2\sqrt{|I_k|}]$. The variables $X_1, \dots, X_r$ live in commuting classically independent subalgebras. In the standard tensor-product model, the spectrum of $\sum_k X_k$ is the Minkowski sum of the individual spectra. Therefore

$$\left\| \sum_{k=1}^r X_k \right\| = 2 \sum_{k=1}^r \sqrt{|I_k|}.$$

Dividing by $\sqrt{L}$ gives the formula for T_K . □

4 Counterexample

Theorem 1. *The Section 6 edge-count monotonicity question in [1] has a negative answer.*

Proof. Let G be the complete 4-partite graph on 10 vertices with part sizes

$$(1, 1, 4, 4),$$

and let H be the complete 4-partite graph on 10 vertices with part sizes

$$(1, 2, 2, 5).$$

Both graphs have clique number 4. The number of edges in a complete multipartite graph with part sizes $n_1, \dots, n_r$ is

$$\frac{1}{2} \left(L^2 - \sum_{k=1}^r n_k^2 \right).$$

For both displayed partitions, $L = 10$ and

$$1^2 + 1^2 + 4^2 + 4^2 = 1^2 + 2^2 + 2^2 + 5^2 = 34.$$

Thus

$$|E(G)| = |E(H)| = \frac{100 - 34}{2} = 33.$$

By Lemma 1,

$$\|T_G\| = \frac{2}{\sqrt{10}}(1 + 1 + 2 + 2) = \frac{12}{\sqrt{10}},$$

whereas

$$\|T_H\| = \frac{2}{\sqrt{10}}(1 + \sqrt{2} + \sqrt{2} + \sqrt{5}) = \frac{2(1 + 2\sqrt{2} + \sqrt{5})}{\sqrt{10}}.$$

These values are not equal: equality would imply $5 = 2\sqrt{2} + \sqrt{5}$, which is impossible since squaring $5 - \sqrt{5} = 2\sqrt{2}$ gives $30 - 10\sqrt{5} = 8$.

Since $|E(G)| = |E(H)|$, the proposed monotonicity would imply both $\|T_G\| \leq \|T_H\|$ and $\|T_H\| \leq \|T_G\|$, hence equality. The strict inequality above contradicts this. Therefore the proposed monotonicity property is false. $\square$

5 Verification Notes

The arithmetic can be checked with:

`python code/verify_partitions.py.`

The script confirms that both graphs have 10 vertices, clique number 4, and 33 edges, and computes the two norm values.

No numerical approximation is used in the proof. The only structural input is the complete multipartite norm formula, proved above and also used in the source paper’s Section 6 discussion.

6 Novelty Check and Limitations

The run’s cheap indexes were searched for the arXiv id, title terms, “trace monoids”, “Turan”, “Khintchine”, and “monoid”. No prior packet for this target was found. Local source inspection showed that the first Turan extremal question is answered in the same paper, while the Section 6 edge-count monotonicity question is posed as unclear. Web searches on 2026-07-03 for exact phrases from the title and question did not reveal a separate later answer.

This counterexample refutes the proposed monotonicity as stated. It does not rule out monotonicity under stronger hypotheses, for example when G is a subgraph of G' , which is already observed in the source, or within a fixed part-size majorization class.

7 Human Review Recommendation

Review as a short counterexample. The main point to check is that the normalization in Lemma 1 matches the source definition of $T_G = (1/\sqrt{L}) \sum_i s_i$.

References

- [1] Patrick Oliveira Santos, Raghavendra Tripathi, and Pierre Youssef. *Khintchine inequalities, trace monoids and Turan-type problems*. arXiv:2506.02517, 2025.